1. RX setup introduction　接收机操作说明

1.1 Dual antenna notes

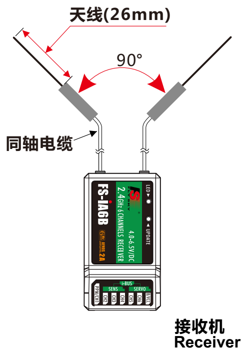
In order to make sure maximum distance between the transmitter and receiver please follow the directions below:

1.The two antennas must be kept as straight as possible.Otherwise,control range will be reduced.
2.The two antennas should be placed at a 90 degree angle to each other,as illustrated in the three pictures below.
3.The antennas must be kept away from conductive materials,such as metal and carbon.A distance of at least 1.5cm is req uired for safeop eration.Conductive materials will not affect the coaxialp art of the antenna,but it is important that the coaxials are not bent to a severe radius.
4.Keep antennas away from the motor,speed controller and other noise souces as much as possible.

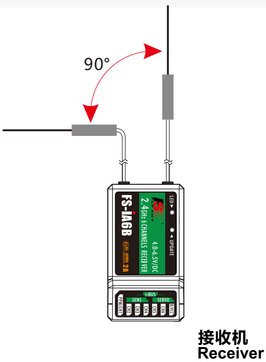
接收机双天线注意事项：

为了让发射及接收距离更远，请注意以下几点：

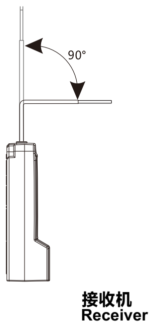
1. 尽量保证双天线笔直，否则将会减小控制范围；
2. 双天线的夹角保持在90°(如图三种方式),这并不是精确的垂直角度，重要的是尽可能保持天线互相远离；
3. 天线应该尽可能远离金属导体，至少要有l.5cm左右的距离。轴电缆段不受此限制，但不要过度弯曲；
4. 尽可能保持天线远离电动机、调速器，和其它的噪声源。



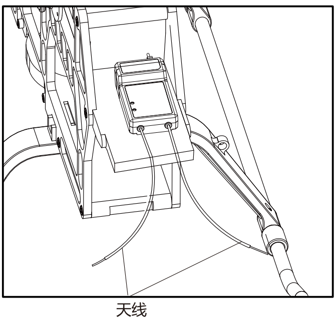
(图 1.1)

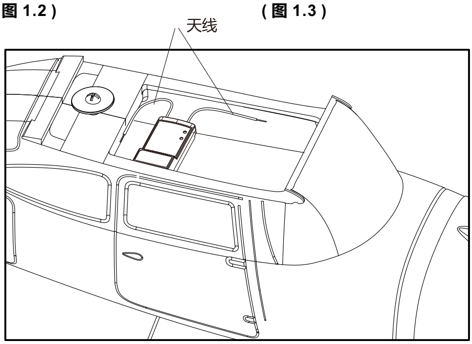


(图 1.2)



(图 1.3)

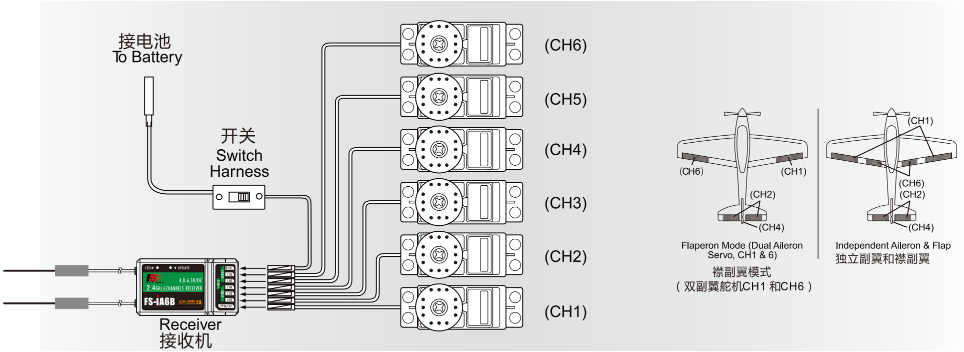




2. Receiver and servo connections 接收机与伺服器连接

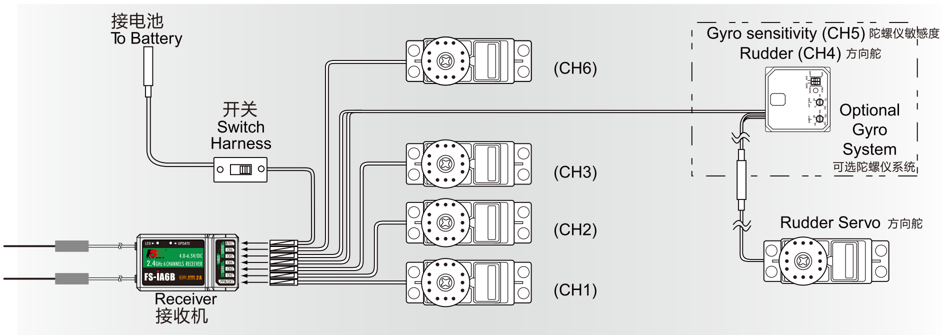
2.1. Receiver and servo connections (aircraft)

飞机模型的接收机与伺服器连接



2.2 Receiver and servo connections(helicopter)

直升机模型的接收机与伺服器连接

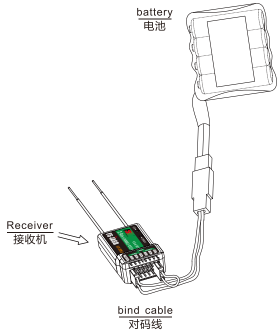


3. 2.4GHz Operation notes 2.4G操作注意事项

3.1 . Binding 对码

The supplied transmitter and receiver are already bound at production time so you don't need to do it. If you are using another transmitter or receiver, you have to first bind them before use as described below:

1. Install batteries in the transmitter and turn it off.
2. Connect the binding jumper to the battery port of the receiver.
3. Connect the battery of the receiver to any channel power supply. The red LED with blink indicating that it is in binding mode.
4. Press and hold the bind key of the transmitter and turn it on.
5. The binding process is finished when the red indicator on receiver flashes more slowly than before. Pull out the binding wire and the red indicator stays on.
6. Disconnect the receiver battery.
7. Turn off then back on the transmitter.
8. Connect all the servos to the receiver and then connect its battery.
9. Check if all servos are working as expected.
10. If anything is wrong, please bind again according aboving rsteps.



对码:

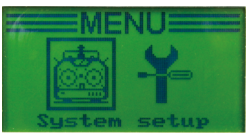
所有遥控产品在出厂的时候都已经对好码，您无需再次对码。如果您需要和其他发射机或接收机对码， 您必须在使用前按照以下方法对码：

1. 将电池装入发射机然后关闭发射机。
2. 将对码线插到接收机电池通道插口。
3. 将接收机电池连接至接收机任意通道，接收机红色指示灯快速闪烁表明处于对码状态。
4. 按住发射机对码按键不松手，同时打开发射机。
5. 接收机红色指示灯由快闪变成慢闪表明对码成功，拔掉对码线，红色指示灯常亮
6. 断开接收机电源。
7. 关闭发射机电源。
8. 将所有舵机连接至接收机,然后就将电池连接到接收机。
9. 检查是否所有的舵机按照要求工作正常。
10. 如果对码失败,请按以上步骤从头再来。

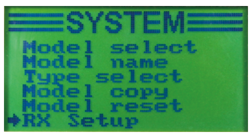
4.Receiver PPM outputs: 接收机PPM输出

From the main menu, select SYSTEM setting option, "RX Setup", "PPM output", and "Receiver" sequentially to turn on PPM output, then CH1 will now output PPM stream conveying 6 signals

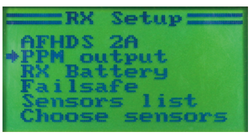
进入主菜单，选择”System”子目录，依次选择“Rx Setup”、“PPM output”子目录，然后选择”Receiver PPM output”为“ON”即可,此时CH1输出为6路PPM信号。



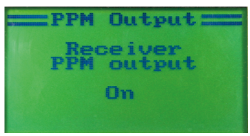
(1)



(2)



(3)

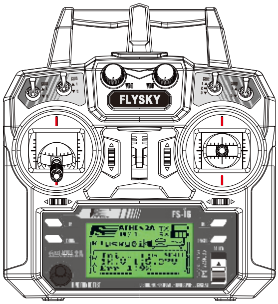


(4)

5.1 Power on 开机

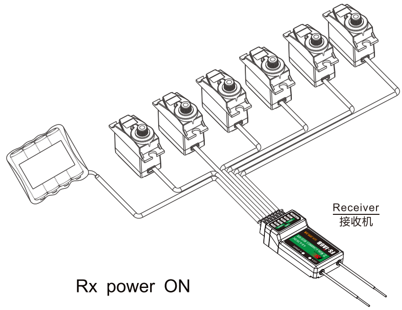
1. Connnect all parts
2. Switch on the transmitter
3. Connect the receiver battery
4. The receiver red LED indicator is solid indicating the presence of a correct signal
5. Use the radio system

1. 连接好所有部件
2. 打开发射机
3. 接通接收机电源
4. 接收机红色指示灯常亮说明信号连接正常.
5 操作系统可以使用



Tx power ON

Power on

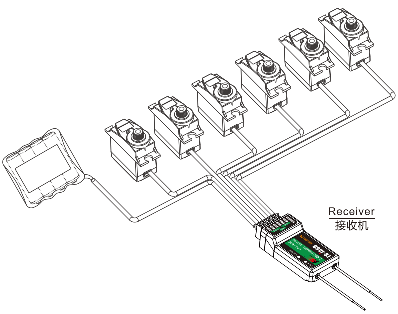


Rx power ON

5.2 Shut down 关机

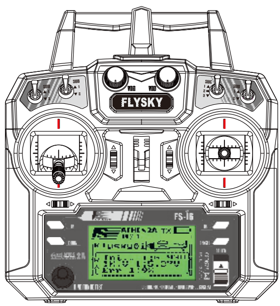
1. Disconnect the receiver battery
2. Switch off the transmitter

1. 断开接收机电源
2. 关闭发射机



Rx power off

Shut down



Tx power off

6. Supporting module 配套模块

6. 01. RPM Telemetry [magnetic] module 磁感应速度采集模块

SPECIFICATIONS :

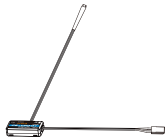
- Model type: car/boat/aircraft
- Monitor range of speed60000RPM
- Power: 4.0-6.5V DC
- Weight: 6.6g
- Size: 31*15*8.5mm
- Color: black

机种参数：

- 适合机种： 车、船、飞机
- 采集速度范围： 0-60000转/分钟
- 输入电源： 4.0-6.5V DC
- 机身重量： 6.6克
- 外形尺寸： 31*15*8.5毫米
- 外观颜色： 黑色

AFHDS 2A

MODEL: FS-CPD01



6. 02. RPM Telemetry [optical] module 光感应速度采集模块

SPECIFICATIONS :

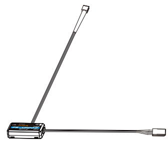
- Model type: car/boat/aircraft
- Monitor range of speed60000RPM
- Power: 4.0-6.5V DC
- Weight: 6.8g
- Size: 31*15*8.5mm
- Color: black

机种参数：

- 适合机种： 车、船、飞机
- 采集速度范围： 0-60000转/分钟
- 输入电源： 4.0-6.5V DC
- 机身重量： 6.8克
- 外形尺寸： 31*15*8.5毫米
- 外观颜色： 黑色

AFHDS 2A

MODEL: FS-CPD02



6. 03. Temperature acquisition module 温度采集模块

SPECIFICATIONS :

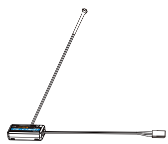
- Model type: car/boat/aircraft
- Monitor range of temperature: -40~250°C
- Power: 4.0-6.5V DC
- Weight: 5.9g
- Size: 31*15*8.5mm
- Color: black

机种参数：

- 适合机种： 车、船、飞机
- 采集温度范围：-40~250度
- 输入电源： 4.0-6.5V DC
- 机身重量： 5.9克
- 外形尺寸： 31*15*8.5毫米
- 外观颜色： 黑色

AFHDS 2A

MODEL: FS-CTM01



6. 04. Voltage acquisition module 电压采集模块

SPECIFICATIONS :

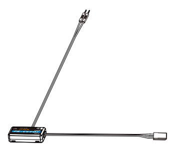
- Model type: car/boat/aircraft
- Monitor range of Voltage: 0-100V DC
- Power: 4.0-6.5V DC
- Weight: 6g
- Size: 31*15*8.5mm
- Color: black

机种参数：

- 适合机种： 车、船、飞机
- 电压采集范围： 0-100V DC
- 输入电源： 4.0-6.5V DC
- 机身重量： 6克
- 外形尺寸： 31*15*8.5毫米
- 外观颜色： 黑色

AFHDS 2A

MODEL: FS-CVT01



6. 05. i-BUS Serial bus receiver i-BUS 串行总线接收机

SPECIFICATIONS :

- Channels: 4
- Model type: car/boat/aircraft
- Weight: 8.1g
- Power: 4.0-6.5V DC
- Size: 30*25.6*13mm
- Color: black
- i-BUS PORT: yes

机种参数：

- 通道个数：4
- 适合机种： 车、船、飞机
- 机身重量： 8.1克
- 输入电源： 4.0-6.5V DC
- 外形尺寸： 30*25.6*13毫米
- 外观颜色： 黑色
- i-BUS 接口： 有

AFHDS 2A

MODEL: FS-CEV04

7. FS-CEV04 i-BUS serial receiver connection　FS-CEV04 i-BUS 串行总线接收机连接说明

串行总线接收机，最多可串联4个模块，共18个通道；按键K1-K4分别对应C1-C4,用于对相应通道的设定；

操作说明：

1、FS-CEV04接收机的“IN”端口对应接收机的“SERVO”端口；
2、FS-CEV04接收机的“OUT”端口，用于串接后级的FS-CEV01接收机，以串联的方式使用。
3、将此总线接收机插入接收机，打开已配对的发射机，接收机电源，LED点亮；
4、操作发射机触控屏，选择接收机设定的主菜单，进入到舵机设定界面；
5、选择需要扩展的通道，此时，总线接收机的LED熄灭；
6、用对码线上的胶针，按下需要的，相应通道的按键，LED自动点亮，表示设定成功；
7、插入舵机，检查设定是否成功；
8、重复以上操作即可完成总线接收机4个通道的设定；
9、当需要更多的通道扩展时，只需要在第一级总线接收机的“SERVO”端口，串接新的总线接收机即可，设定的操作方法相同。

注意：当总线接收机的负载过重，电流较大时，请将主接收机的电源分支出来并联接入，单独供电加大负载的能力，否则可能会因电流过大，烧坏串联的线材。

Serial bus receiver can connect 4 modules with 18 channels in serial at most. Button K1 and K4 correspond to C1 and C4 respectively.

Operation:

1. “IN” port of FS-CEV04 receiver corresponds to “SERVO” port of receiver.
2. The “OUT” port of FS-CEVO4 receiver is used to connect post level FS-CEV04 receiver。
3. Insert the bus receiver to receiver, and then switch on the matched transmitter and receiver. The LED will be on.
4. Select main menu of receiver setup to enter the interface of servo setup.
5. Select channel which need to be expanded, meanwhile LED of bus receiver is off.
6. Push relevant channel button by plastic needle of matching line. The setup is successful if LED flashes automatically.
7. Insert servo to check.
8. Set up 4 channels of bus receiver as above steps.
9. Just connect a new bus receiver with “SERVO” port of first stage bus receiver if more channel needed. Set up the new one as above steps.

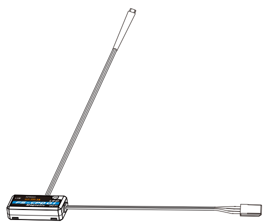
Notice:

when the load of serial bus receiver is excessive and electric current is higher than usual, please supply power directly to the serial bus receiver or it will break cables.

8. Data telemetry connection　数据采集模块连接

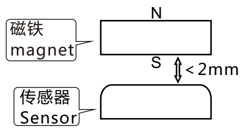
8.1 采集模块的操作使用说明：

FS-CPD01：磁感应转速采集模块



操作使用说明：

1、将所配的3PIN插头，一端插入速度采集模块的“OUT”位置，另一端插入 接收机的“SENS”位置或接另外的感应器的“IN”位置，如上图所示；
2、将图3的传感器放在磁铁的旁边，磁铁固定在需要测试的轴向转动的地方。如：模型车的轮毂内侧，如下图所示，传感器与磁铁相距两毫米以内，磁铁的南极或北极与传感器保持平行。
3、打开发射机，接收机电源，在显示屏的接收机窗口内，会发现并显示 “Motor speed 2：0RPM”,试着转动轮子，转速的值会发生变化，则表示安装成功。



Data telemetry operation instruction

FS-CPD01: revolving speed module.

Operation:

1. Insert one end of standard 3 PIN plug into “OUT” port of speed acquisition module, and insert the other end into “SENS” port of receiver or other sensor, as picture above.
2. Forexample：Insidehub of the model, the distance between sensor and magnet is less than2mm. The north Pole or the south pole of themagnet has to be paralleled with sensor.
3. Switch on transmitter and receiver. “Motor speed 2:0RPM” will be shown in receiver window in display screen. Speed value changes as turning wheel, which means installation is successful.

采集模块的操作使用说明：

FS-CPD02：光感应转速采集模块

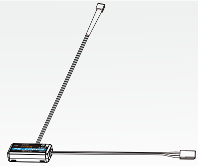
操作使用说明：

1、将所配的3PIN插头，一端插入速度采集模块的“OUT”位置，另一端插入接收机的“SENS”位置或接另外的感应器的“IN”位置，如上图所示；

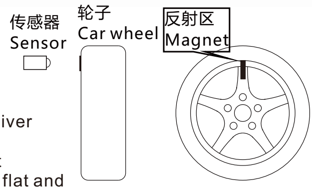
2、将图2所示，传感器与反射贴纸固定在轮子的侧面平面上，保持贴纸平整，并与传感器垂直；（备注：贴纸与轮子的颜色反差要大）传感器和贴纸距离要保持适中。

3、打开发射机，接收机电源，在显示屏的接收机窗口内，会发现并显示“Motor speed 2：0RPM”,试着转动轮子，转速的值会发生变化，则表示安装成功。

备注: 也可安装在模型车的从动齿轮上,采用相同的方法采集齿轮的转速.



8.2 Telemetry module



FS-CPD02: optical rotation speed telemetry module

Operation:

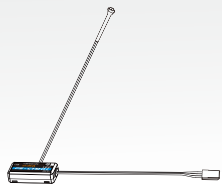
1. Connect one end of the standard 3 PIN plug to the "out" port of the speed telemetry module and the other end to the "SENS" port of the receiver or the previous sensors “in” port as pictured above.
2. As picture 2 shows, affix the sensor and the reflection decals on the flat surface of the side of any rotating part (gear, car wheel...). Keep decals flat and perpendicular to the sensor. (Remark: high color contrast between decals and rotating part gives better result). Maintain sufficient safety distance between the sensor and the decals to avoid any damage.
3. Switch on the transmitter and the receiver. “Motor speed 2: 0RPM” will be displayed in the main screen. The speed displayed will follow the speed of the rotating part monitored by the rotation speed sensor, indicating a successful installation.

Remark: You can also fix it to the driven gear of the model car. Use the same method to collect RPM data of gear.

FS-CTM01：温度采集模块连接

操作使用说明：

1. 将所配的3PIN连接线，一端插入温度采集模块的“OUT”位置，另一端插入接收机的“SENS”位置或接另外的感应器的“IN”位置；
2. 将温度的传感器本体，使用海棉双面贴粘在适当的位置（如：马达，电池本体上），并与被测试物表面紧贴；
3. 打开发射机，接收机电源，在显示屏的接收机窗口内，会发现并显示“Temperature 1：25.0℃”，表示安装成功，25.0℃即为采集到的温度数据。



8.3 FS-CTM01: Temperature telemetry connection

Operation:

1. Insert one end of standard 3 PIN plug into “OUT” port of temperature module, and insert the other end into “SENS” port of receiver or other sensor, as picture above.
2. Adhere temperature sensor to proper place (such as motor and battery) tightly by sponge double stick.
3. Switch on transmitter and receiver. “Temperature 1:25.0℃” will be shown in receiver window in display screen, which means installation is successful, and 25.0℃ is the temperature collected.